

Tutorial Week 1

1. The mass of air in a room $3\text{ m} \times 4\text{ m} \times 25\text{ m}$ is known to be 350 kg. Determine the density, specific volume, and specific weight. [Hint: $v = \rho^{-1}$, $\gamma = \rho g$]

2. Complete the following if $g = 9.81\text{ m/s}^2$ and $V = 10^4\text{ L}$.

	$v(\text{m}^3/\text{kg})$	$\rho(\text{kg}/\text{m}^3)$	$\gamma(\text{N}/\text{m}^3)$	$m(\text{kg})$	$W(\text{N})$
(a)	20.00				
(b)		2.00			
(c)			4.00		
(d)				100.00	
(e)					100.00

3. Determine the weight of a mass of 12 kg at a location where the acceleration of gravity is 9.77 m/s^2 .

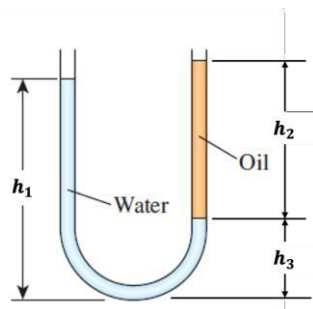
4. Calculate the force necessary to accelerate a 8000-kg rocket vertically upward at the rate of 28 m/s^2 . [Hint: The total upward force from the engine must accomplish two things, consider drawing a free-body diagram to visualize the forces.]

5. A 2200-kg automobile traveling at 90 kph hits the rear of a stationary, 1000-kg automobile. After the collision, the large automobile slows to 60 kph, and the smaller vehicle has a speed of 78 kph. What has been the increase in internal energy, taking both vehicles as the system? [Hint: The increase in internal energy is the kinetic energy that was **lost** during the collision, $E_{k1} + U_1 = E_{k2} + U_2$.]

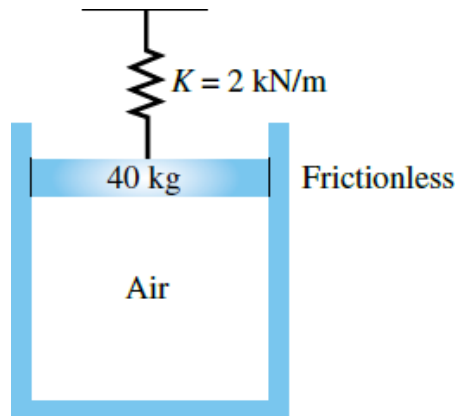
6. Calculate the force due to the pressure acting on the 1.2-m diameter horizontal hatch of a submarine submerged 800 m below the surface. [Hint: Use $\rho_{\text{water}} = 1000\text{ kg/m}^3$ and the total force is the pressure at that depth multiplied by the area of the hatch]

7. A U-tube with arms open to the atmosphere. Water is poured into the U-tube from one arm, and light oil from the other. One arm contains water, while the other arm contains both fluids. Determine the height of each fluid in that arm.

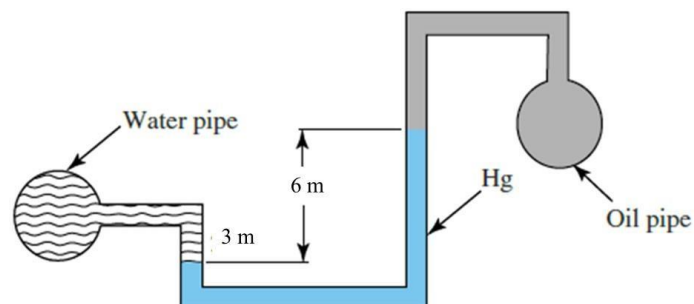
Given: $\rho_{\text{oil}} = 790\text{ kg/m}^3$ $h_1 = 50\text{ cm}$ $h_3 = \frac{1}{4}h_2$ [Hint: Apply the principle of hydrostatic equilibrium at a specific horizontal level]



8. Calculate the pressure in the 150-mm diameter cylinder. The spring is compressed 35 cm. [Hint: Force balance]



9. Determine the pressure difference between the water pipe and the oil pipe. The top of the mercury column is at the same elevation as the middle of the oil pipe. [Hint: Pressure balance]



10. A horizontal 2.5-m-diameter gate is located in the bottom of a water tank. Determine the force F required to just open the gate. [Hint: To open the gate, the moment created by the applied force must be at least equal to the moment due to water pressure.]

